

TOPIC 5

Parkinson's vs Healthy — Biomarker Comparison

5 Charts | 5 KPIs | 5 Key Questions

Complete Step-by-Step Power BI Guide | All DAX Formulas

PADS Smartwatch Dataset | 469 Patients | Datathon Presentation

Chart 1 Radar Chart	Chart 2 Clustered Bar	Chart 3 Donut Chart	Chart 4 Scatter Plot	Chart 5 Heatmap
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TOPIC 5 OVERVIEW

Complete Chart Plan — All 5 Unique Chart Types

Topic 5 is the grand conclusion of your Parkinson's Disease analysis. It combines all four data tables to create a comprehensive biomarker comparison between PD and healthy patients. Every chart uses a completely unique type — no two charts are the same. Each chart has one most important KPI and one most important analysis question.

Chart	Type	Most Important Question	Unique Purpose
Chart 1	Radar Chart	Does sensor pattern differ between Left and Right Wrist in PD?	Full 6-axis movement profile shape
Chart 2	Clustered Bar	How many more symptoms do PD patients carry vs Healthy?	Side-by-side symptom burden
Chart 3	Donut Chart	Is the PD biomarker slice disproportionately large?	Combined biomarker proportion
Chart 4	Scatter Plot	Do PD patients with more symptoms also show higher tremor?	3-variable cluster analysis
Chart 5	Heatmap	Which activity causes most movement difficulty for PD vs Healthy?	Tremor grid across 11 activities

ALL DAX FORMULAS

Create Every Item Below Before Building Any Chart

IMPORTANT: Follow this exact order — Calculated Columns first, then Measures. Do NOT skip any item. Every formula below is used by at least one chart.

SECTION 1 — Calculated Columns

How to create: Go to Data View > Click the table name on the right > Table Tools (top menu) > New Column > Paste the formula > Press Enter

1.1 Age Group Column — Add to Dim_Patients

What it does: Groups each patient's age into 3 categories — Under 50, 50 to 65, Over 65. Used as a slicer on all pages to filter by age group.

DAX — New Column in Dim_Patients

```
Age_Group =  
SWITCH(  
TRUE(),  
Dim_Patients[age] < 50, "Under 50",  
Dim_Patients[age] >= 50 && Dim_Patients[age] <= 65, "50 to 65",  
Dim_Patients[age] > 65, "Over 65",  
"Unknown"  
)
```

1.2 Observation_Key Column — Add to Dim_Observation

What it does: Creates a unique key by combining Patient ID + Record Name + Device Location. This key is used to connect Dim_Observation to Fact_Timeseries with a One-to-Many relationship in the schema.

DAX — New Column in Dim_Observation

```
Observation_Key =  
Dim_Observation[Patient ID]  
& "_"  
& Dim_Observation[Record Name]  
& "_"  
& Dim_Observation[Device Location]
```

1.3 Observation_Key Column — Add to Fact_Timeseries

What it does: Creates the matching key in Fact_Timeseries. Must be identical in format to the key created in 1.2 above. These two keys are what Power BI uses to join the tables.

DAX — New Column in Fact_Timeseries

```

Observation_Key =
Fact_Timeseries[Patient ID]
& "_"
& Fact_Timeseries[Record Name]
& "_"
& Fact_Timeseries[Device Location]

```

1.4 Symptom Count Column — Add to Dim_Patients

What it does: Counts how many True (yes) answers each patient gave in the questionnaire. A patient with 10 True answers has a Symptom_Count of 10. This becomes the Symptom Biomarker score used in Charts 2, 3 and 4.

DAX — New Column in Dim_Patients

```

Symptom_Count =
CALCULATE(
COUNTROWS(
FILTER(
Dim_Questionnaire,
Dim_Questionnaire[Answer] = TRUE()
)
),
FILTER(
Dim_Questionnaire,
Dim_Questionnaire[Patient ID] = Dim_Patients[id]
)
)

```

1.5 Movement Score Normalised — Add to Fact_Timeseries

What it does: Converts raw accelerometer readings into a 0 to 100 scale. The lowest tremor reading = 0 and the highest = 100. This allows movement and symptoms to be combined into one score in Chart 3.

DAX — New Column in Fact_Timeseries

```

Movement_Score_Normalised =
VAR MinVal = MINX(ALL(Fact_Timeseries), Fact_Timeseries[Accel_X_Mean])
VAR MaxVal = MAXX(ALL(Fact_Timeseries), Fact_Timeseries[Accel_X_Mean])
RETURN
DIVIDE(
Fact_Timeseries[Accel_X_Mean] - MinVal,
MaxVal - MinVal,
0
) * 100

```

SECTION 2 — Measures

How to create: Go to Home tab > New Measure > Paste the formula > Press Enter. Measures can be created in any table. They recalculate automatically whenever a slicer is applied on the report page.

2.1 Average Tremor — PD and Healthy

What they do: Calculate the average accelerometer reading separately for PD patients and for healthy patients. These power the KPI cards and Chart 1.

DAX — Create Both Measures

```
Avg_Tremor_PD =  
CALCULATE(  
    AVERAGE(Fact_Timeseries[Accel_X_Mean]),  
    Dim_Patients[condition] = "Parkinson's Disease"  
)  
  
Avg_Tremor_Healthy =  
CALCULATE(  
    AVERAGE(Fact_Timeseries[Accel_X_Mean]),  
    Dim_Patients[condition] = "Healthy"  
)
```

2.2 Average Symptom Count — PD and Healthy

What they do: Calculate the average number of symptoms per patient for each group. These power the Clustered Bar Chart (Chart 2) and KPI cards.

DAX — Create Both Measures

```
Avg_Symptoms_PD =  
CALCULATE(  
    AVERAGE(Dim_Patients[Symptom_Count]),  
    Dim_Patients[condition] = "Parkinson's Disease"  
)  
  
Avg_Symptoms_Healthy =  
CALCULATE(  
    AVERAGE(Dim_Patients[Symptom_Count]),  
    Dim_Patients[condition] = "Healthy"  
)
```

2.3 Difference Measures — PD minus Healthy

What they do: Calculate the exact gap between PD and Healthy for tremor and symptoms. These are the most impactful KPI card numbers — a positive number means PD is higher.

DAX — Create Both Measures

```
Tremor_Difference =  
[Avg_Tremor_PD] - [Avg_Tremor_Healthy]
```

```
Symptom_Difference =  
[Avg_Symptoms_PD] - [Avg_Symptoms_Healthy]
```

2.4 Total Patient Count — PD and Healthy

What they do: Count the total number of patients in each group. Used in the KPI cards at the top of the page.

DAX — Create Both Measures

```
Total_PD_Patients =  
CALCULATE(  
COUNTROWS(Dim_Patients),  
Dim_Patients[condition] = "Parkinson's Disease"  
)  
  
Total_Healthy_Patients =  
CALCULATE(  
COUNTROWS(Dim_Patients),  
Dim_Patients[condition] = "Healthy"  
)
```

2.5 Combined Biomarker Score

What it does: Adds the normalised movement score and normalised symptom score together into one single 0 to 200 score per condition. Higher score = higher Parkinson's biomarker load. Powers the Donut Chart (Chart 3).

DAX — New Measure

```
Combined_Biomarker_Score =  
VAR AvgMovement = AVERAGE(Fact_Timeseries[Movement_Score_Normalised])  
VAR AvgSymptom = DIVIDE(  
AVERAGE(Dim_Patients[Symptom_Count]),  
30,  
0  
) * 100  
RETURN AvgMovement + AvgSymptom
```

2.6 Average Tremor Per Activity — for Heatmap

What it does: Calculates average accelerometer reading per activity session per condition. Each cell in the Heatmap (Chart 5) gets its colour from this measure.

DAX — New Measure

```
Avg_Tremor_Per_Activity =  
CALCULATE(  
AVERAGE(Fact_Timeseries[Accel_X_Mean]),  
ALLEXCEPT(  
Fact_Timeseries,  
Fact_Timeseries[Record Name],
```

```
Dim_Patients[condition]
)
)
```

2.7 Left Wrist and Right Wrist Tremor — for Radar Chart

What they do: Calculate average tremor separately for Left Wrist and Right Wrist in PD patients. These power the Left vs Right wrist comparison in Chart 1.

DAX — Create Both Measures

```
Avg_Tremor_LeftWrist_PD =
CALCULATE(
    AVERAGE(Fact_Timeseries[Accel_X_Mean]),
    Dim_Patients[condition] = "Parkinson's Disease",
    Dim_Observation[Device Location] = "LeftWrist"
)

Avg_Tremor_RightWrist_PD =
CALCULATE(
    AVERAGE(Fact_Timeseries[Accel_X_Mean]),
    Dim_Patients[condition] = "Parkinson's Disease",
    Dim_Observation[Device Location] = "RightWrist"
)
```

2.8 Radar Chart Axis Measures — 12 Total

What they do: Power the 6 axes of the Radar Chart. Create 6 measures for PD and 6 for Healthy — one per sensor axis (Accel X, Y, Z and Gyro X, Y, Z).

DAX — Create 12 Measures Using This Pattern

```
-- PD Measures (6 total):
Avg_Accel_X_PD = CALCULATE(AVERAGE(Fact_Timeseries[Accel_X_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")
Avg_Accel_Y_PD = CALCULATE(AVERAGE(Fact_Timeseries[Accel_Y_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")
Avg_Accel_Z_PD = CALCULATE(AVERAGE(Fact_Timeseries[Accel_Z_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")
Avg_Gyro_X_PD = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_X_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")
Avg_Gyro_Y_PD = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_Y_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")
Avg_Gyro_Z_PD = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_Z_Mean]),
    Dim_Patients[condition]="Parkinson's Disease")

-- Healthy Measures (6 total) – repeat above replacing condition with "Healthy":
Avg_Accel_X_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Accel_X_Mean]),
    Dim_Patients[condition]="Healthy")
Avg_Accel_Y_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Accel_Y_Mean]),
    Dim_Patients[condition]="Healthy")
```

```
Avg_Accel_Z_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Accel_Z_Mean]),  
Dim_Patients[condition]="Healthy")  
  
Avg_Gyro_X_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_X_Mean]),  
Dim_Patients[condition]="Healthy")  
  
Avg_Gyro_Y_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_Y_Mean]),  
Dim_Patients[condition]="Healthy")  
  
Avg_Gyro_Z_Healthy = CALCULATE(AVERAGE(Fact_Timeseries[Gyro_Z_Mean]),  
Dim_Patients[condition]="Healthy")
```


KPI CARDS

5 Headline Cards — Top Row of the Report Page

PURPOSE: 5 KPI cards placed in a single row across the top of the page. Judges read these in 5 seconds and understand the entire story before looking at any chart. Always build these first before adding any chart.

Card	Title	DAX Measure	Background Colour	Expected Value
1	Total PD Patients	[Total_PD_Patients]	Light Red (#FADBD8)	~235 patients
2	Total Healthy Patients	[Total_Healthy_Patients]	Light Blue (#D6E8F7)	~234 patients
3	Tremor Difference	[Tremor_Difference]	Light Orange (#FDEBD0)	Positive number
4	Symptom Difference	[Symptom_Difference]	Light Purple (#E8DAEF)	Positive number
5	Avg Tremor — PD	[Avg_Tremor_PD]	Light Red (#FADBD8)	Higher than Healthy

Step-by-Step — How to Build KPI Cards

STEP 1

Add New Report Page

Right-click on the pages panel at the bottom of Power BI > Add Page > Double-click the page name and rename it 'Topic 5 Biomarker Comparison'

STEP 2

Add 5 Card Visuals

Go to Visualizations pane > Click the Card visual icon > Add 5 Card visuals to the canvas. Arrange them in a single row across the full top of the page. Make all 5 cards equal width

STEP 3

Assign Measures to Cards

Click Card 1 > In the Fields pane on the right drag [Total_PD_Patients] into the Fields box. Repeat for each card using the measures from the table above

STEP 4

Set Card Titles

For each card go to Format pane (paint roller icon) > Title > On > Type the title from the table above > Set font to Bold size 12

STEP 5

Apply Background Colours

For each card go to Format pane > Background > On > Click the colour picker > Enter the hex colour from the table above. PD cards = red tones, Healthy = blue tones

STEP 6

Add Border and Shadow

Format pane > Border = On for all cards. Shadow = On. This makes the KPI row stand out cleanly from the charts below it

CHART 1

Left Wrist vs Right Wrist Movement Profile in PD (Radar Chart)

WHAT IT SHOWS: A spider/radar chart comparing the sensor movement profile between Left Wrist and Right Wrist in Parkinson's Disease patients across all 6 sensor axes. PD typically starts on one side of the body — if one wrist shows a larger polygon, it confirms the asymmetric nature of Parkinson's Disease. This is clinically unique and no other team will likely analyse this.

MOST IMPORTANT KPI

Avg Tremor — Left Wrist vs Right Wrist (PD)

DAX Measure: [Avg_Tremor_LeftWrist_PD] vs [Avg_Tremor_RightWrist_PD]

Shows which wrist has higher average tremor in PD patients. If Left Wrist > Right Wrist, PD started on the left side in most patients. This asymmetry is a key clinical characteristic of Parkinson's Disease.

Step-by-Step — How to Build Radar Chart

STEP 1

Install Radar Chart Visual

Go to Visualizations pane > Click the three dots icon (...) at the bottom > Click 'Get more visuals' > In the search box type 'Radar Chart' > Find the one by xViz or MAQ Software > Click Add > It will now appear in your Visualizations pane

STEP 2

Add Radar Chart to Canvas

Click the Radar Chart icon in Visualizations pane > It appears on the canvas > Drag to position below the KPI cards on the left side > Resize to approximately half the page width

STEP 3

Add the 12 Axis Measures to Values

In the Fields pane drag all 12 radar measures into the Values field of the Radar Chart: Avg_Accel_X_PD, Avg_Accel_Y_PD, Avg_Accel_Z_PD, Avg_Gyro_X_PD, Avg_Gyro_Y_PD, Avg_Gyro_Z_PD and all 6 Healthy equivalents

STEP 4

Add Device Location to Legend

Drag Dim_Observation[Device Location] into the Legend field. This creates two polygons automatically — one for LeftWrist and one for RightWrist in PD patients

STEP 5

Format Colours and Labels

Click Format pane (paint roller) > Data Colors > Set LeftWrist = Dark Blue (#1B3A6B), RightWrist = Light Blue (#2E6DA4). Turn on Data Labels to show values on each axis point. Set Title = 'Left vs Right Wrist Movement Profile — PD Patients'

STEP 6

Add KPI Card Beside Chart

Add a small Card visual beside the radar chart > Drag [Avg_Tremor_LeftWrist_PD] into it. Add another Card with [Avg_Tremor_RightWrist_PD]. These two numbers show the wrist asymmetry instantly

RECOMMENDED CHART SETTINGS: Use Device Location as the Legend field (not condition) to show Left vs Right wrist. Apply a Condition slicer filtered to 'Parkinson's Disease only' so the chart shows only PD patients. Filter by 'Relaxed' session in the Record Name slicer to show resting tremor asymmetry which is the most clinically meaningful comparison.

MOST IMPORTANT QUESTION

Does the overall sensor movement pattern differ significantly between Left Wrist and Right Wrist in Parkinson's Disease patients — indicating asymmetric tremor?

HOW TO ANSWER

Look at the two polygons on the radar chart. If the LeftWrist polygon is noticeably larger or differently shaped than the RightWrist polygon, it confirms that PD tremor is asymmetric — affecting one side of the body more than the other. This is a clinically important finding because Parkinson's Disease almost always starts on one side of the body before spreading. Say: 'Our data confirms the asymmetric nature of PD — the [Left/Right] wrist shows consistently higher sensor readings, indicating this is the dominant side of disease onset in our patient group.'

CHART 2

Symptom Burden PD vs Healthy (Clustered Bar Chart)

WHAT IT SHOWS: Side-by-side bars comparing the average number of non-motor symptoms reported by PD patients versus healthy patients. The taller PD bar shows that Parkinson's patients silently carry more symptoms — including sleep disorders, depression, cognitive issues and digestive problems. Simple, clear and immediately understandable to any judge.

MOST IMPORTANT KPI

Avg Symptom Count — PD vs Healthy

DAX Measure: [Avg_Symptoms_PD] vs [Avg_Symptoms_Healthy]

Shows average number of symptoms per patient in each group. The gap between these two numbers is your non-motor biomarker headline finding. A bigger gap = stronger evidence that PD creates a non-motor burden.

Step-by-Step — How to Build Clustered Bar Chart

STEP 1

Add Clustered Bar Chart

Visualizations pane > Click the Clustered Bar Chart icon (horizontal bars side by side) > It appears on canvas > Position it top-right below the KPI cards row, next to the Radar Chart

STEP 2

Set the X-Axis (Categories)

In the Fields pane drag Dim_Patients[condition] into the X-axis field. This puts 'Parkinson's Disease' and 'Healthy' as the two bars

STEP 3

Set the Y-Axis (Values)

Drag [Avg_Symptoms_PD] into the Y-axis Values field. Then also drag [Avg_Symptoms_Healthy] into the same Values field. Power BI will show both as grouped bars

STEP 4

Turn on Data Labels

Click Format pane (paint roller icon) > Data Labels > On. This shows the exact average symptom number on top of each bar — makes the difference immediately readable without needing to read the axis

STEP 5

Format Bar Colours and Reference Line

Format pane > Data Colors > Parkinson's Disease bar = Red (#E74C3C), Healthy bar = Blue (#2E6DA4). Then go to Analytics pane > Constant Line > Add line at the overall average. Label it 'Overall Average'

STEP 6

Set Title and Final Formatting

Format pane > Title = 'Average Non-Motor Symptom Count: PD vs Healthy' > Font 12pt Bold > Set background to white > Add border. Turn off gridlines for cleaner look

RECOMMENDED CHART SETTINGS: Sort the bars in descending order so PD appears first. Add a subtitle below the chart title saying 'Out of 30 total questionnaire symptoms' to give context to the numbers. Enable tooltips to show the exact average on hover. Keep Y-axis starting at 0 — never truncate the axis as it misleads judges.

MOST IMPORTANT QUESTION	How many more non-motor symptoms do PD patients report on average compared to healthy patients — and is this difference clinically significant?
HOW TO ANSWER	Read the data labels on both bars. Subtract the Healthy number from the PD number. If PD = 8 and Healthy = 3, the difference is 5 symptoms — a 167% higher burden. To check clinical significance: divide the difference by 30 (total questions). If the result is 10% or more, it is clinically meaningful. Say: 'PD patients report X more symptoms on average than healthy patients — a Y% higher symptom burden. This non-motor signature is present in our data alongside the movement biomarker, confirming PD affects multiple body systems.'

CHART 3

Combined Biomarker Score (Donut Chart)

WHAT IT SHOWS: A donut chart showing what proportion of the total combined biomarker score (movement + symptoms normalised to 0-200) belongs to PD patients vs Healthy patients. Even though PD patients are approximately 50% of the study population, they should take a much larger share of the combined biomarker load — this disproportionate share is your strongest evidence for multi-modal detection.

MOST IMPORTANT KPI

PD Share % of Combined Biomarker Score

DAX Measure: [Combined_Biomarker_Score] split by condition

The percentage of the total combined biomarker load carried by PD patients. If PD patients are 50% of the population but carry 65%+ of the score, they have a disproportionately higher biomarker burden — key detection evidence.

Step-by-Step — How to Build Donut Chart

**STE
P 1**

Add Donut Chart

Visualizations pane > Click the Donut Chart icon (circle with hole) > It appears on canvas > Position it in the bottom-left area below the Radar Chart

**STE
P 2**

Set Legend and Values

Drag Dim_Patients[condition] into the Legend field. Drag [Combined_Biomarker_Score] into the Values field. Power BI automatically splits the donut into PD slice and Healthy slice

**STE
P 3**

Turn on Detail Labels

Format pane > Detail Labels > On > Label Style = select 'Category and Percentage' OR 'All detail labels' > This shows both the condition name and the percentage on each slice

**STE
P 4**

Format Slice Colours

Format pane > Data Colors > Parkinson's Disease slice = Red (#E74C3C) > Healthy slice = Blue (#2E6DA4). Consistent with all other charts in the topic

**STE
P 5**

Set Inner Radius and Centre Label

Format pane > Shapes > Inner Radius = 60% > This creates the donut hole. Go to Insert > Text Box > Place it in the centre of the donut > Type 'Biomarker Load' in the text box > Font 10pt bold dark blue

**STE
P 6**

Set Title and Format

Format pane > Title = 'Combined Biomarker Score Distribution: Movement + Symptoms' > Font 12pt Bold Teal > Add border > Turn on Legend at bottom of chart

RECOMMENDED CHART SETTINGS: Add a tooltip that shows the exact Combined_Biomarker_Score value on hover for each slice. Place a small explanatory text box below the donut saying: Score = Normalised Movement (0-100) + Normalised Symptom Count (0-100). This helps judges understand what the combined score represents without needing explanation.

**MOST
IMPORTANT
QUESTION**

What percentage of the total combined biomarker score does the PD slice represent — and is this disproportionately large compared to the PD population share?

**HOW TO
ANSWER**

Read the percentage label on the PD slice. Then compare it to the PD population share (approximately 50% from the KPI cards). If PD is 50% of patients but takes 65% of the combined score, PD patients carry 30% more biomarker burden per person than expected. Say: 'PD patients represent approximately 50% of our study population but account for X% of the total combined biomarker load. This disproportionate share proves that combining movement sensor data with symptom questionnaire responses amplifies the Parkinson's detection signal significantly beyond what either source shows alone.'

CHART 4

Symptom Burden vs Tremor Intensity (Scatter Plot)

WHAT IT SHOWS: A scatter plot with symptom count on X-axis, tremor intensity on Y-axis, bubble size = age, colour = condition (PD red, Healthy blue). If PD patients with more symptoms also show higher tremor, it proves that movement and symptom biomarkers are connected — PD attacks multiple body systems simultaneously. This is a unique multi-modal finding only possible because this dataset combines both smartwatch sensor data and clinical questionnaires.

MOST IMPORTANT KPI

Avg Age at Diagnosis — PD Patients

DAX Measure: CALCULATE(AVERAGE(Dim_Patients[age_at_diagnosis]),
Dim_Patients[condition]="Parkinson's Disease")

Shows the typical age when PD patients in this study were diagnosed. Combined with the scatter trend, shows whether earlier diagnosis correlates with lower symptom and tremor burden.

Additional DAX Measure Needed for This Chart

DAX — New Measure for KPI Card

```
Avg_Age_At_Diagnosis_PD =  
CALCULATE(  
  AVERAGE(Dim_Patients[age_at_diagnosis]),  
  Dim_Patients[condition] = "Parkinson's Disease"  
)
```

Step-by-Step — How to Build Scatter Plot

STEP 1

Add Scatter Chart

Visualizations pane > Click the Scatter Chart icon (dots scattered) > It appears on canvas > Position it bottom-right below the Clustered Bar Chart

STEP 2

Set X-Axis to Symptom Count

In Fields pane drag Dim_Patients[Symptom_Count] into the X-axis field. This puts symptom count on the horizontal axis — higher = more symptoms

STEP 3

Set Y-Axis to Tremor

Drag Fact_Timeseries[Accel_X_Mean] into the Y-axis field > Set aggregation to Average. This puts tremor intensity on the vertical axis — higher = more tremor

STEP 4	Set Details for One Bubble Per Patient Drag Dim_Patients[id] into the Details field. This creates one bubble per patient instead of one bubble per condition
STEP 5	Set Bubble Size and Colour Size field = Dim_Patients[age] (larger bubble = older patient). Legend field = Dim_Patients[condition]. Format > Data Colors: PD = Red (#E74C3C), Healthy = Blue (#2E6DA4)
STEP 6	Add Trend Line and Title Analytics pane > Trend Line > Apply for each group separately. A rising trend among red PD bubbles proves the symptom-tremor connection. Title = 'Symptom Count vs Tremor Intensity (Bubble Size = Age)'

RECOMMENDED CHART SETTINGS: Enable Tooltips to show Patient ID, Symptom Count, Tremor and Age on hover. Add a reference line on both axes at the overall average to create 4 quadrants. PD patients should cluster in the top-right quadrant (high symptoms + high tremor). Healthy patients should cluster in the bottom-left quadrant (low symptoms + low tremor). This quadrant separation is your key visual finding.

MOST IMPORTANT QUESTION	Do PD patients with more symptoms (higher X-axis) also show higher tremor intensity (higher Y-axis) — proving that movement and symptom biomarkers are connected in Parkinson's Disease?
HOW TO ANSWER	Look at the red PD bubbles in the scatter plot. If the bubbles in the right side of the chart (high symptom count) also sit higher on the Y-axis (high tremor), it confirms the connection. Check the PD trend line — an upward slope proves the positive correlation. Say: 'PD patients with more non-motor symptoms also show higher tremor readings — confirming that Parkinson's Disease is a multi-system condition. Our combined biomarker approach captures both dimensions simultaneously, creating a stronger detection signal than either sensor data or questionnaire alone.'

CHART 5

Tremor Intensity Across All Activities (Heatmap)

WHAT IT SHOWS: A grid where rows = 11 activity sessions, columns = condition (PD vs Healthy), and cell colour = average tremor intensity. Dark red cells = high tremor. Light blue cells = low tremor. This shows which physical activities produce the most movement difficulty for PD patients compared to healthy individuals — proving that Parkinson's tremor is present across ALL tasks, not just resting states.

MOST IMPORTANT KPI

Highest Tremor Activity — PD Patients

DAX Measure: [Avg_Tremor_Per_Activity] — filtered to PD, sorted descending

The activity session with the highest average tremor reading for PD patients. This is the single most impactful physical task for detecting PD tremor in this dataset. The darker the heatmap cell for this activity, the stronger the finding.

Step-by-Step — How to Build Heatmap Using Matrix Visual

STEP 1

Add Matrix Visual

Visualizations pane > Click the Matrix icon (grid of squares) > It appears on canvas > Position it at the bottom of the page spanning the full width. This is Power BI's built-in matrix that we convert into a heatmap using conditional formatting

STEP 2

Set Rows to Activity Sessions

In Fields pane drag Fact_Timeseries[Record Name] into the Rows field. This puts all 11 activity sessions (Relaxed, CrossArms, TouchNose etc.) as the rows of the matrix

STEP 3

Set Columns to Condition

Drag Dim_Patients[condition] into the Columns field. This creates two columns — one for Parkinson's Disease and one for Healthy

STEP 4

Set Values to Avg Tremor Per Activity

Drag [Avg_Tremor_Per_Activity] into the Values field. Each cell now shows the average tremor reading for that activity and that condition

STEP 5

Apply Conditional Formatting — THE KEY STEP THAT CREATES THE HEATMAP

Click the dropdown arrow next to [Avg_Tremor_Per_Activity] in the Values field > Conditional Formatting > Background Color > Format Style = Gradient > Minimum Color = Light Blue (#D6EAF8) > Maximum Color = Dark Red (#922B21) > Click OK. The matrix is now a heatmap — dark red = high tremor, light blue = low tremor

STE P 6

Sort and Final Formatting

Click the Rows header > Sort Descending by [Avg_Tremor_Per_Activity] > This puts the highest tremor activities at the top. Turn off Row Subtotals and Column Subtotals in Format pane. Set Title = 'Tremor Intensity Heatmap: All Activities vs Condition'. Turn on cell Values to show numbers inside each coloured cell

RECOMMENDED CHART SETTINGS: After applying conditional formatting, check that the colour gradient looks correct — the darkest cell should be the highest tremor value. If colours look wrong, go back to Conditional Formatting and verify Min/Max values. Add a legend text box below the heatmap saying: Dark Red = High Tremor | Light Blue = Low Tremor. This helps judges read the chart without explanation. Turn off the row and column grand totals as they distract from the individual cells.

MOST IMPORTANT QUESTION

Which physical activity causes the most movement difficulty for PD patients compared to healthy patients — and is the PD column consistently darker across all 11 activities?

HOW TO ANSWER

Look at the PD column — identify the darkest red cell. That activity is where PD patients show the highest tremor and therefore the most movement difficulty. Then scan the entire PD column vs the Healthy column row by row. If every PD cell is darker than its corresponding Healthy cell, it proves Parkinson's tremor affects every physical task — not just rest. Say: 'The [activity name] session produces the highest tremor in PD patients. More importantly, the PD column is consistently darker than the Healthy column across all 11 activities — proving that Parkinson's Disease creates a pervasive movement impairment that affects every physical task our patients performed.'

COMPLETE SUMMARY

5 Charts — 5 KPIs — 5 Questions — Zero Overlaps

Chart	Type	KPI	Most Important Question
Chart 1	Radar	Left vs Right Wrist Tremor	Does sensor pattern differ Left vs Right Wrist in PD?
Chart 2	Clustered Bar	Avg Symptom Count PD vs Healthy	How many more symptoms do PD patients carry?
Chart 3	Donut	PD % of Combined Biomarker Score	Is PD biomarker slice disproportionately large?
Chart 4	Scatter Plot	Avg Age at Diagnosis — PD	Do more symptoms = higher tremor in PD patients?
Chart 5	Heatmap	Highest Tremor Activity — PD	Which activity causes most difficulty for PD?

DAX Checklist — Tick Before Building

Done	Item	Table	Used In
☐	Age_Group column	Dim_Patients	Slicer on all pages
☐	Observation_Key — Dim_Observation	Dim_Observation	Schema relationship
☐	Observation_Key — Fact_Timeseries	Fact_Timeseries	Schema relationship
☐	Symptom_Count column	Dim_Patients	Charts 2, 3, 4
☐	Movement_Score_Normalised	Fact_Timeseries	Chart 3 Donut
☐	Avg_Tremor_PD + Healthy	Any table	KPI + Charts 1, 2
☐	Avg_Symptoms_PD + Healthy	Any table	KPI + Chart 2
☐	Tremor_Difference	Any table	KPI Card 3
☐	Symptom_Difference	Any table	KPI Card 4
☐	Total_PD + Healthy Patients	Any table	KPI Cards 1 + 2
☐	Combined_Biomarker_Score	Any table	Chart 3 Donut
☐	Avg_Tremor_Per_Activity	Any table	Chart 5 Heatmap
☐	Avg_Tremor_LeftWrist_PD	Any table	Chart 1 Radar
☐	Avg_Tremor_RightWrist_PD	Any table	Chart 1 Radar
☐	Avg_Age_At_Diagnosis_PD	Any table	Chart 4 KPI

[]	12 Radar axis measures	Any table	Chart 1 Radar
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FINAL DATATHON TIP: Build in this exact order: DAX first > KPI Cards > Chart 1 > Chart 2 > Chart 3 > Chart 4 > Chart 5 > Add all 4 Slicers. Test every slicer after adding it. Use Red = PD and Blue = Healthy consistently across every single chart. Come back for the Presentation Script when all charts are built!